

Showing Work with a Calculator: Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit in each work box show the operation(s) that you entered into the calculator and the result.

1. Aurora plays the trombone in her high school band. The table shows her practice times for one week.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Minutes	20	22	18	24	22	48	56

- a. Determine the range.

Range	38
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- b. Determine the mode.

Mode	22
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- c. Determine the mean.

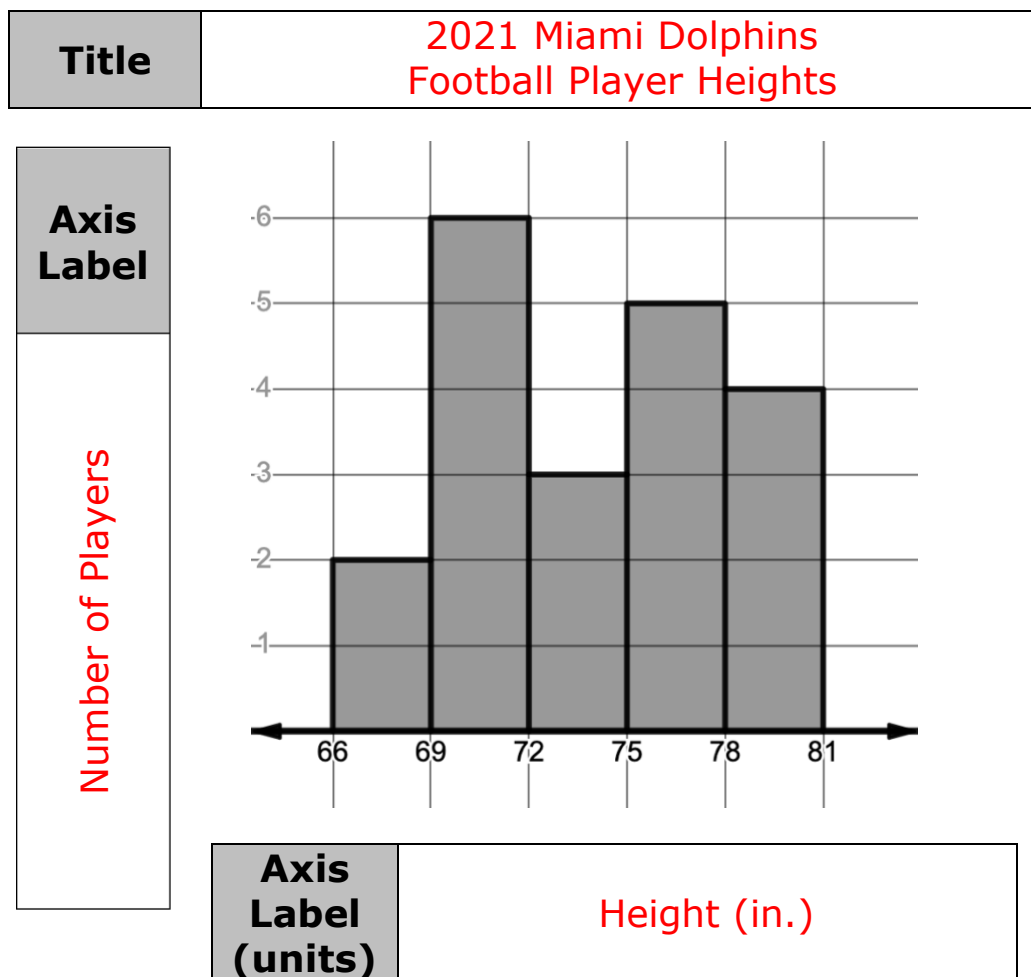
work	
Mean	30

2. The height, in inches (in.), of some of the Miami Dolphins football players during the 2021 season is shown.

71, 71, 69, 70, 70, 78, 77, 77, 77, 77, 75, 78, 68, 79, 78, 68, 73, 71, 73, 73

An incomplete histogram of the data is shown. Complete the histogram by providing:

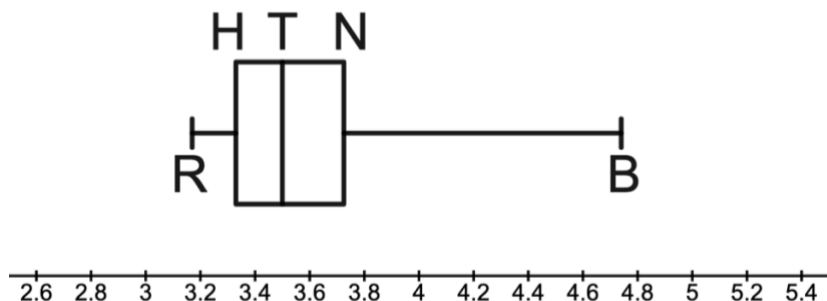
- a title
- a label for the vertical axis
- a label for the horizontal axis.



3. The US Bureau of Labor Statistics tracks the price of a pound of chocolate chip cookies. Some of the prices are shown.

{\\$3.19, \\$3.50, \\$3.17, \\$3.50, \\$3.48, \\$3.55, \\$3.47, \\$3.90, \\$4.74}

The box plot of the data is shown.



Select all of the statements that are true.

- ☐ 10% of the data are between H and T .
- ☒ 25% of the data are between T and N .
- ☒ 25% of the data are between R and H .
- ☒ 50% of the data are between H and N .
- ☐ 50% of the data are between R and N .
- ☐ 75% of the data are between N and B .

4. The table shows the question five different students wrote.

Student	Question
Erika	How many students are in the cafeteria today?
Jamal	How many times do students get their hair cut each year?
Nina	What is the average height of the sixth graders at our school?
Robert	What is your favorite club at the school?
Tiana	What is for lunch today?

**Student could also correctly complete the statement saying that Erika did not write a statistical question.*

Complete the statement.

- ☐ Erika
- ☐ Jamal
- ☐ Nina
- ☐ Robert
- ☒ Tiana

did not write a statistical question because

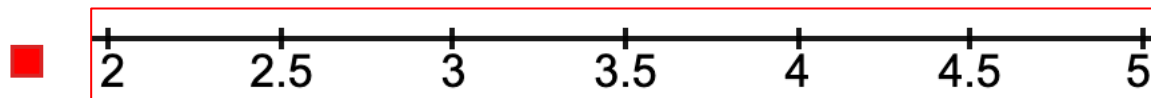
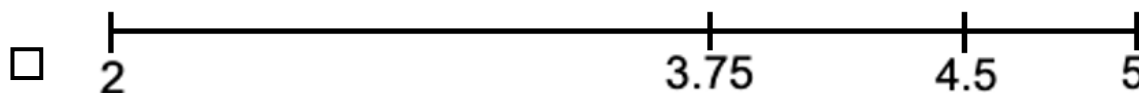
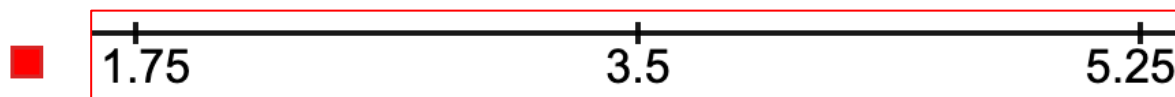
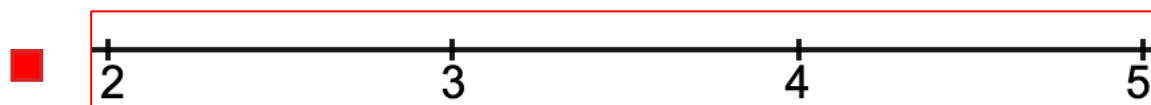
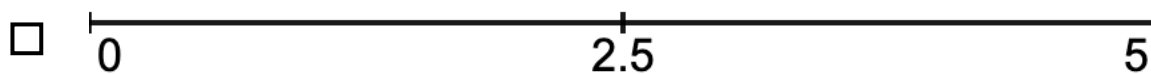
the question

- ☐ is not numerical.
- ☐ is not categorical.
- ☒ does not have variability.

5. Kadeesha designs clothes. The table shows the amount of fabric, in yards (yd), she needs to buy.

Design Name	Fabric (yd)			
	Fabric A	Fabric B	Fabric C	Fabric D
Caheerah	$3\frac{1}{4}$	$2\frac{3}{4}$	$3\frac{1}{2}$	$4\frac{1}{8}$
Hania	$4\frac{1}{8}$	$3\frac{1}{2}$	$2\frac{1}{4}$	$2\frac{3}{4}$
Pallaton	$3\frac{3}{4}$	$4\frac{1}{8}$	$4\frac{3}{4}$	$3\frac{5}{8}$
Reilyyn	$3\frac{1}{2}$	$2\frac{3}{8}$	$4\frac{1}{8}$	$3\frac{1}{2}$

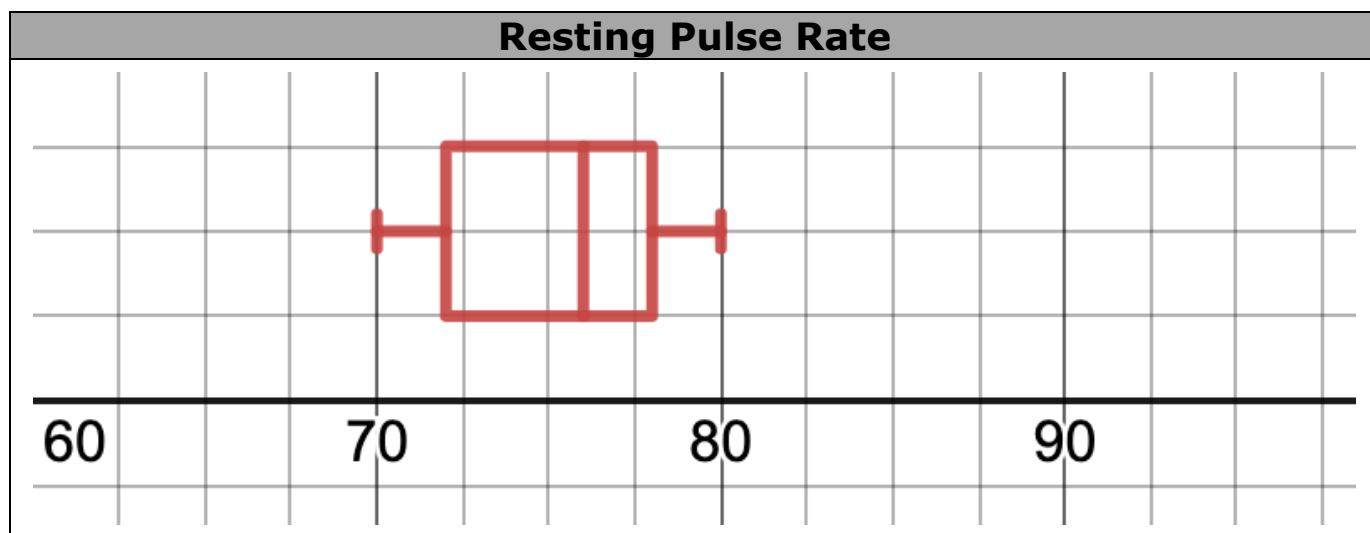
Kadeesha would like to create a histogram of fabric amounts. Select all of the intervals for the bins that are reasonable.



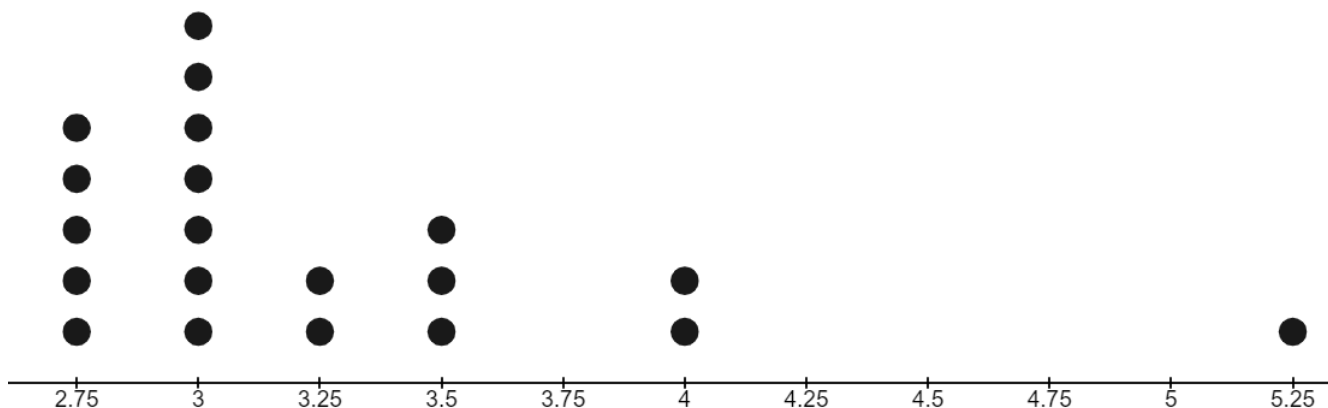
6. Terri recorded her resting pulse rate at different times of the day. Her data is shown and is labeled as data set A .

$$A = \{70, 70, 70, 70, 72, 72, 72, 73, 76, 76, 77, 77, 78, 78, 78, 80, 80, 80\}$$

Create a box plot.



7. The cost, in dollars, of a healthy meal in 20 different countries is shown in the dot plot (line plot).

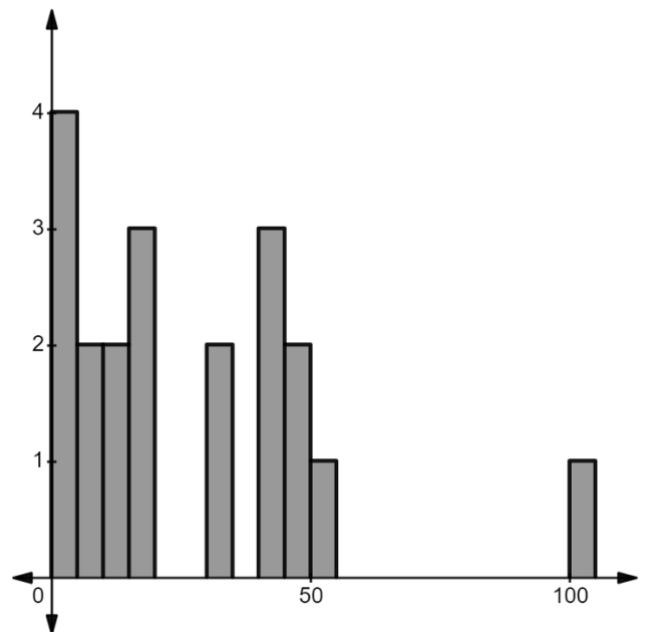


Which statistical question may result in the data shown in the dot plot (line plot)?

- A. What is the cost, in dollars, of a healthy meal?
- B. How many countries have a healthy meal that costs \$3.50?
- C. Which countries have healthy meals that cost less than \$5.50?
- D. How much, per person, does it cost to make a meal in the United States?

8. Sixth graders were asked for the number of text messages they received on a particular day. A histogram representing the number of text messages a sample of 20 sixth graders received in one day is shown.

Number of Text Messages			
0	7	3	46
10	34	17	12
32	0	51	42
40	15	101	2
42	47	8	18



a. Complete the statements.

The shape of the distribution of received text messages can be best

described as

- ☒ skewed,
- ☐ uniform,
- ☐ symmetric,

with a

- ☐ gap
- ☐ peak
- ☒ cluster

between 40 and

55 text messages. This distribution has a

- ☒ gap
- ☐ peak
- ☐ cluster

between

20 and 30 text messages and an outlier between

- ☐ 30 and 35
- ☐ 55 and 100
- ☒ 100 and 105

text messages.

b. Determine the median.

work	
Median	17.5